Sharon Wong

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Education

Purdue University 2025
1st year PhD. Biomedical Engineering *West Lafayette, IN*
Earlham College 2021-2025
Bachelor of Arts in Neuroscience and Computer Science (GPA: 3.96 / 4.00) *Richmond, IN*

Lab & Technical Skills

Programming & Software: Python, R, Matlab, C, SQL, React.js, HTML/CSS, Markdown, Git
Machine Learning & Modeling: YOLOv5/YOLO8/YOLO11, CNNs, Random Forest, SVM, XGBoost, SHAP, LIME, TensorFlow, Keras, ImageDataGenerator, Scikit-learn
Quantitative Methods: Monte Carlo simulations, heuristic search, control systems modeling, signal processing, parameter optimization, electrochemical modeling, statistical analysis
Data Visualization & Tools: Tableau, Seaborn, Matplotlib, Ipycytoscape, EEG-MNES.py, PsychoPy
Wet Lab & Biomedical: EEG, eye-tracking, rat/mouse behavioral work, Electrochemical Impedance Spectroscopy (EIS) and Cyclic Voltammetry (CV), Nova Autolab, microtomy, Leica microscopy, force sensors, soldering

Research Experience

Research Assistant | *Skills: Electrochemical impedance spectroscopy, Nova Autolab,* Aug 2025 – Present
Weldon School of Biomedical Engineering, Purdue University, West Lafayette, IN ***Under Dr. Kevin Otto***

- Building quantitative models to interpret impedance spectra, automate data extraction, and identify predictive signatures of electrode degradation.
- Building Python pipelines that stream and synchronize data from Autolab, generator waveform, and oscilloscopes, automatically saving, organizing, and preprocessing impedance time series for downstream modeling.
- Developing feature extraction and prediction models that link impedance signatures to recording quality and simulate decision rules for safe, closed loop rejuvenation under electrochemical and hardware limits.

Research Assistant | *Skills: Ipycytoscape, Monte Carlo simulation.* Aug 2022 – May 2025
Physics, Earlham College, Richmond, IN ***Under Dr. Michael Lerner***

- Designed and implemented Monte Carlo simulations to model Twist1-associated gene networks and identify non-obvious drug targets.
- Developed node-weighted “walker” models to simulate biological signal diffusion and quantify prioritization of transcriptional vs. protein-protein interactions.
- Ran large scale parameter sweeps and sensitivity analyses on diffusion models to quantify how network topology and edge weights shift drug target rankings and robustness.

Quantitative Modeling of Alzheimer’s Detection via MRI | Skills:YOLO11, CNNs, TensorFlow May 2025
Computer Science, Earlham College, Richmond, IN

- Developed an end-to-end deep learning pipeline to classify Alzheimer’s disease using MRI data, achieving 98% confidence in binary classification using YOLO models.
- Compared model performance across traditional and deep learning methods, using F1-score, precision/recall, and transfer learning for optimization.
- Proposed model integration into clinical decision systems and presented interpretability strategies to increase transparency in diagnostic pipelines.

Decision-Making Under Uncertainty in Medical Reports | Skills: Quantitative analysis May 2025
Neuroscience, Earlham College, Richmond, IN

- Conducted a behavioral study evaluating how information presentation (text-only vs. visuals) affects patient comprehension, decision accuracy, and confidence in clinical choices.
- Used inferential statistics (t-tests, chi-square) to analyze data from 115 participants and explored links between cognitive outcomes and neural pathways (mPFC, LIP).
- Investigated implications for patient-centered design and decision-support tools, with applications in medical informatics and neuroeconomics.

Research Assistant Jan 2024 – July 2024
Neuroscience, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA ***Under Dr. Michael Platt***

(1) **Pitches Entrepreneurship Project** | *Skills: B-Alert EEG System, PsychoPy, Tensorflow*

- Implemented synchronization between B-Alert EEG systems and PsychoPy software, optimizing data integrity and event marking for neuromarketing research.
- Analyzed EEG frontal alpha asymmetry (FAA) across multiple subjects and videos to derive insights into cognitive responses.

- Developed a Machine Learning model with TensorFlow achieving 97% accuracy in predicting venture capital funding outcomes based on EEG frontal alpha asymmetry features.
- Analyzed time-series signal data to correlate neural engagement metrics with financial success.

(2) **Fox Company Project** | *Skill: Cogwear EEG, Eye-tracking, OpenBCI (software) EEG-MNES.py.*

- Collaborated with the Wharton Neuroscience Initiative and Fox Media Company to apply EEG and eye-tracking technology for analyzing viewer engagement and predicting attention spikes in advertising contexts.
- Assisted in collecting participant data and organized eye-tracking and clinical system data collection for over 200 participants.

Research Assistant | *Skills: Arduino, OnShape, Soldering, Force sensory mechanisms.* May – Aug 2023
Biomedical Engineering, Case Western Reserve University, Cleveland, OH ***Under Dr. Dustin Tyler***

- Collaborated with Veterans Affairs on neuroengineering research to innovate in human fusion technology.
- Evaluated and contrasted two Arduino-based force sensory mechanisms for accurate force localization, incorporating CAD design, soldering, and Arduino setup in experimental designs.
- Conducted research on varied force levels to overcome challenges in receptive sensory fields and sensory thresholds, enhancing sensory device accuracy and contributing to advancements in sensory prosthetics for veterans.

Research Assistant | *Skills: Slurm, Beam Search.* Jan – May 2023
Computer Science, Earlham College, Richmond, IN ***Under Dr. Sofia Lemon***

- Analyzed Anytime Weight A* in Korf's puzzles domain to learn efficient limited memory heuristic search planning.
- Emphasized lower weight of the cost, resulting in longer search time, lower cost, higher quality, and lower coverage.

Research Assistant | *Skills: MATLAB, handling Rats.* May – July 2022
Neuroscience, Indiana University School of Medicine, Indianapolis, IN ***Under Dr. Yadav P. Amol***

- Trained 10+ rats with spinal cord electrons to complete operating and discrimination tasks in behavior boxes.
- Discovered rats' capability to differentiate artificial sensations during varying spinal cord stimulations via behavior tasks.

Research Assistant | *Skills: Microtome, Microscope.* Aug 2020 – June 2021
Institute of Brain Science, National Yang-Ming Chiao Tung University, Taipei, Taiwan ***Under Dr. Shih-Chieh Lin***

- Supported lab with mice brain preparation, including perfusion, sectioning, embedding, and microtomy operations.
- Managed Leica microscope setup for basal forebrain histology analysis.

Publication

Yun, J. H., Heaton, S., **Wong, J. H.**, Klein, P., & Platt, M. L. *(under review). Brain signals from new ventures predict entrepreneur fundraising success.* PLOS ONE

Pham, B., **Wong, J. H.**, Kim, S., Yin, Y., & Skiena, S. (2025). *Word definitions from large language models.* Accepted to the IEEE International Conference on Semantic Computing (ICSC). https://doi.org/10.48550/arXiv.2311.06362

Poster Presentation

Azzouz A, ***Wong, J. H****, Perrone M.C., Bader J.S., Lerner M.G* (2025, Feb 18). Prioritizing Cancer Drug Targets with Monte Carlo Simulations of Network Diffusion [Poster presentation]. *Biophysics Society 2025*, Los Angeles Convention Center, Los Angeles, CA.
Wong, J. H*., Jin, H. Y., & Michael L. Platt* (2024, Oct 7). Predicting entrepreneurial pitch success using behavioral, textual, and EEG neuro forecasting [Poster presentation]. *Society of Neuroscience 2024*, McCormick Place Convention Center, Chicago, IL.
Wong, J. H*., McGann, L., Jakes, R., & Tyler, D.* (2023, July 26). Exploring the Effect of Contact Location on FSR and Piezoelectric Sensor Output to Transduce Rapidly- and Slowly-Adapting Tactile Feedback for Sensory Prostheses [Poster presentation]. *APT Symposium 2023*, Case Western Reserve University, Cleveland, OH.
Biomedical Engineering Society. (2023, October 13). Seattle Convention Center, Seattle, WA.
Midwestern Psychological Association. (2024, April 18). Palmer House Hilton Hotel, Chicago, IL.

Wong, J. H*., & Yadav, A. P.* (2022, July 26). Studying discrimination of artificial sensations induced by aperiodic spinal cord stimulation in rodents [Poster presentation]. *STARK Neuroscience Symposium 2022*, Indianapolis University School of Medicine, Indianapolis, IN.

Leadership Experience

Campus Lead, Google Developer Student Club July 2023 – July 2025
Earlham College, Richmond, IN

- Invited to Google I/O and NA Connection to explore the depth of Gemini 1.5 Pro and flash multimodal case use.
- Implemented strategic marketing initiatives, achieved a total membership increase of 61 members, reflecting a 15% growth, and enhanced member registration rates by 108%.

Department Representative Aug 2023 – May 2025
Computer Science, Earlham College, Richmond, IN

- Facilitated effective communication between students and the department, enhancing cooperation and engagement.

Professional Experience

Hall Director Aug 2024 – May 2025
Office of Residence Life, Earlham College, Richmond, IN
- Supervised and mentored a team of 5 Resident Assistants to manage a living-learning community of 70 residents, facilitating high-level conflict mediation and operational logistics.
- Led crisis response protocols for high-pressure emergency situations, coordinating efficiently with campus safety and administration to ensure rapid resolution.

Teacher Assistant in PSYC 245: Research Methods and Statistics Jan 2023– May 2025
Department of Psychology, Earlham College, Richmond, IN ***Under Dr. Kyle***
- Mentored students in statistical programming (R, Java), shifting focus from rote grading to interactive code debugging and logic troubleshooting.
- Facilitated collaborative problem-solving sessions to help students translate theoretical statistical concepts into executable code for data analysis

Resident Assistant Aug 2022 – Dec 2023
Office of Residence Life, Earlham College, Richmond, IN
- Enforced college policies to ensure a safe and healthy living environment for 30+ undergraduate students.
- Planned and executed community events aimed at promoting social interactions.

Pre-Health Peer Mentor Aug 2022 – May 2024
Center of Global Health, Earlham College, Richmond, IN
- Orchestrated events for 2023 National Health Professions Week, contributing to the execution cycle by demonstrating leadership abilities, connecting with the local health-focused community of 30+ individuals, and fostering cross-functional relationships within the organization.

Competition

Project Manager & Front-End Developer Oct 20–23, 2023
HarvardHacks 2023, Boston, MA
Project: M2-OOD – Mental Health Tracker for Neurodivergent Users
- Led a cross-disciplinary team to design a real-time wearable mental wellness tracker integrating GPT-based conversational agents for cognitive check-ins.
- Developed responsive front-end interfaces using React.js and styled-components with seamless back-end API integration.
- Applied AI techniques for emotional state classification and participated in product pitching, reaching finalist stage among 200+ teams.

Project Manager & Front-End Developer Apr 21–23, 2023
LA Hacks 2023, Los Angeles, CA
Project: E3 – Easy Efficient Education Platform
- Built an adaptive learning app that tailors educational resources based on users' cognitive learning styles (visual, auditory, kinesthetic).
- Managed Agile-style sprint planning, oversaw UI prototyping in Figma, and implemented real-time content delivery via Firebase.
- Presented a functional MVP with backend + AI logic; recognized with Honorable Mention out of 100+ submissions for innovation in edtech.

Honors & Awards

Phi Beta Kappa (ΦΒΚ) July. 2024
An exclusive honor society that extends membership invitations to only the top 10% of liberal arts and sciences graduates at institutions.

Russell M. Lawall Award in Biomedical Research April. 2025
This award is presented to students selected by the Health Service Program Committee to encourage and facilitate research with a faculty member in the broad areas of health and medicine.

William Roha Computer Science Endowed Award April. 2025
To be awarded annually to seniors in recognition of outstanding work in computer science.

Anthony Maggard Neuroscience Award April. 2025
To be awarded annually to seniors in recognition of outstanding work in neuroscience

Psi Chi (ΨΧ) July. 2024
The International Honor Society in Psychology, recognizing academic excellence and fostering professional growth within the discipline.

Charles A. Frueauff Foundation Award. April. 2024
To be awarded to junior mathematics or science majors with high academic achievement, with special emphasis on accomplishments in science, and evidence of creativity.

Epic Advantage Independent Student Research Mar. 2024
Center for Entrepreneurship, Innovation, and Creativity Grant Oct. 2023
Center for Global Health Yunger Fellowship May. 2023
Eli Lilly Discovery Day March. 2023
Davis United World Colleges Scholar Aug. 2021